Hit Song Prediction Across Genres: A Domain-Shift and Feature Importance Analysis Using Spotify Audio Features

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***Abstract*—This paper investigates whether a universal formula exists for predicting hit songs from audio features, or whether musical success is a genre-conditioned phenomenon that resists a single global model. We treat musical genres as distinct machine learning domains and conduct six experiments on the Spotify Tracks Dataset, comprising 113,999 tracks across 114 genres. Using per-genre L2-regularized logistic regression as a baseline, we construct a 114×114 cross-genre F1 matrix and measure a mean generalization gap of 0.114, present in 110 of 114 genres, confirming widespread domain shift. L1 regularization reveals no universally predictive feature; valence and danceability are the most consistent, while energy and key are highly genre-specific. Random Forest models agree on the weakest features but diverge on the strongest, and achieve a mean F1 of only 0.221 versus the logistic regression baseline of 0.467. PCA underperforms the baseline in 110 genres, while PLS is more competitive but does not consistently surpass the full-feature model. A correlation of $r = 0.093$ between shared tracks and cross-genre performance, combined with 3,332 tracks carrying conflicting hit labels across genres, confirms that the observed shift reflects concept drift rather than distributional differences alone. Our results show that hit-song prediction is fundamentally genre-conditioned, and that no compact set of audio features reliably predicts popularity across all genres.**

***Index Terms*—hit-song prediction, domain shift, audio features, logistic regression, random forest, feature importance**

I. Introduction

What makes a song a hit? This question has fascinated music producers, record labels, and researchers for decades. In the era of digital streaming, the problem has gained a new empirical dimension: platforms such as Spotify now provide richly quantified representations of tracks through continuous-valued acoustic descriptors capturing rhythm, timbre, energy, and mood, alongside detailed listener engagement statistics. This infrastructure has transformed hit-song prediction from a speculative art into a tractable machine learning problem, giving rise to a growing body of work commonly referred to as *hit-song science* [1].

Despite steady methodological progress, from support vector machines on hand-crafted features [1] to deep convolutional networks on mel-spectrograms [2], the field still lacks a clear answer to its central question. Most studies train a single model on a pooled, genre-mixed dataset and evaluate it on a random held-out split from the same distribution [3, 4], implicitly assuming that the relationship between audio features and popularity is consistent across genres. This evaluation protocol is optimistic by design: it ensures that the genre composition of the test set mirrors the training set, masking any generalization penalty that would arise in real-world deployment when a model trained predominantly on pop is applied to jazz or classical tracks. Raza and Nanath [5] concluded there is no universal formula for a hit song, and Sebastian Jung and Mayer [6] found genre to be the strongest predictor of popularity, yet neither study directly measured cross-genre generalization. Earlier work by Fan and Casey [7] showed that models trained on one music market fail to transfer to another, and Salganik et al. [8] established a theoretical ceiling by demonstrating that popularity is partly driven by social influence rather than audio characteristics alone. These findings collectively suggest that genre-level domain shift may be a central, underexplored factor in hit-song prediction.

This paper addresses that gap by treating musical genres as distinct machine learning domains and conducting a systematic domain-shift analysis across 114 genres of the Spotify Tracks Dataset. We train L2-regularized logistic regression classifiers independently per genre, evaluate each model across all genres to measure the cross-genre performance gap, analyze how feature importance changes across domains using both L1 logistic regression coefficients and Random Forest Gini impurity scores, and test whether PCA and Partial Least Squares (PLS) can produce representations that generalize better across genre boundaries. The scale of the analysis, 114 genres and a 114×114 pairwise F1 matrix, allows us to identify not only the overall magnitude of domain shift but also which genres transfer well to others, which features drive prediction consistently, and where dimensionality reduction helps or hurts. This investigation is guided by five research questions:

- **RQ1:** Do models trained on within-genre audio features generalize to other genres, and how large is the domain-shift gap?
- **RQ2:** Does feature importance remain stable across genres, or does it vary substantially by domain?

- **RQ3:** Can L1 regularization identify a compact set of domain-invariant features?
- **RQ4:** Do Random Forest importances support L1 feature rankings, and does a nonlinear ensemble improve over logistic regression?
- **RQ5:** Do PCA and PLS pipelines improve cross-genre generalization over the full-feature L2 baseline?

Our results show that domain shift is present in 110 of 114 genres, that no audio feature is universally predictive across all genres, and that neither PCA nor PLS consistently surpasses the full-feature baseline. Our contributions are threefold: a 114×114 F1 matrix quantifying pairwise cross-genre generalization; an analysis of domain-invariant versus genre-specific audio features across both logistic regression and Random Forest models; and a benchmarking of PCA and PLS against the full-feature baseline across all 114 genres.

II. Motivation

Playlist placement on algorithmic feeds such as Spotify's Discover Weekly or Release Radar can determine whether a track reaches millions of listeners or remains largely unnoticed. Record labels, independent artists, and streaming platforms all have strong incentives to understand which audio characteristics predict engagement and whether those characteristics are universal or dependent on genre.

Most existing hit-song prediction studies train a single classifier on mixed-genre datasets and evaluate it using random train-test splits from the same distribution [3, 4]. This setup hides the performance degradation that may occur when a model trained on one genre, such as pop, is applied to another genre, such as jazz or classical music. Several high-performing studies also focus heavily on commercially dominant genres, making genre-stratified evaluation necessary to determine whether strong predictive performance generalizes broadly or remains genre-specific [9].

In this context, domain shift is not simply a statistical issue but a meaningful cultural phenomenon. Musical genres differ intentionally in rhythm, instrumentation, structure, and emotional tone. Features that indicate popularity in one genre may have little predictive value in another. This perspective also offers an alternative interpretation of one of the most influential negative findings in the literature. Pachet and Roy [10] reported that global acoustic features could not reliably predict popularity across 32,000 mixed-genre tracks. Rather than showing that audio features are inherently weak predictors, this result may indicate that meaningful within-genre patterns are diluted when all genres are pooled together into a single model.

Several recent studies move toward this perspective but stop short of directly investigating it. Sebastian Jung and Mayer [6] identified genre as the strongest predictor of popularity but did not measure cross-genre generalization gaps. Middlebrook and Sheik [3] did not analyze genre-specific feature importance. Raza and Nanath [5] concluded that there is no universal formula for a hit song without examining whether this conclusion arises from cross-genre heterogeneity or from the inherent unpredictability of popularity identified by Salganik et al. [8]. This paper addresses these gaps simultaneously through large-scale cross-genre evaluation, feature-importance analysis, and dimensionality-reduction experiments.

III. Background

A. Hit-Song Science

Dhanaraj and Logan [1] introduced *hit-song science* by training SVMs and AdaBoost on acoustic and lyric features to predict chart success above chance, framing popularity prediction as a supervised binary classification problem that remains the standard setup today. Pachet and Roy [10] challenged this at scale, finding that global acoustic features across 32,000 mixed-genre tracks yielded only weak baselines. This result can now be interpreted as a consequence of pooling heterogeneous genres into a single model, which is precisely the limitation this paper investigates. Ni et al. [11] partially reconciled the two positions by showing accuracy gains on UK Top-40 data and documenting that feature importance shifts across decades, providing an early instance of temporal domain shift directly analogous to the cross-genre shift studied here. More recently, deep learning approaches have pushed accuracy higher [2, 12], but cross-genre generalization remains largely unexplored.

A key theoretical constraint comes from Salganik et al. [8], who showed in a controlled experiment that identical songs achieved dramatically different commercial outcomes across independent listener populations due to social-influence cascades. This implies that popularity is partly self-reinforcing through cumulative advantage rather than a deterministic function of audio quality, establishing an upper bound on what any audio-feature model can achieve regardless of its complexity. Understanding how much of the remaining explainable variance is genre-specific versus universal is one of the central questions this paper addresses.

B. Domain Shift in Machine Learning

Domain shift occurs when the joint distribution of inputs and labels differs between training and deployment. Two forms are particularly relevant to this work. *Covariate shift* occurs when $p(X)$ changes but $p(Y|X)$ stays constant, meaning genres occupy different regions of the feature space while preserving the same feature-to-popularity mapping. *Concept drift* is the stronger case where $p(Y|X)$ itself changes across domains, such as when high danceability strongly predicts popularity in electronic music but carries little signal in folk or classical. Concept drift is more damaging to cross-genre transfer because a model can no longer rely on any consistent relationship between features and labels. Distinguishing between these two forms of shift is one goal of the coefficient and label-agreement analysis in Experiments 2 and 3.

Transfer learning addresses domain shift by adapting knowledge learned in a source domain to a target domain. Choi et al. [13] demonstrated that CNN features trained on music tagging transfer effectively across MIR classification and regression tasks, establishing the canonical reference for audio domain adaptation. Hung et al. [14] extended this to low-resource genre classification through pre-trained speech-model reprogramming, showing that domain adaptation remains viable even with limited target-domain data. Although this paper does not implement transfer learning, the pairwise generalization gaps measured in Experiment 2 directly quantify the domain shift that such methods would need to overcome, and can serve as a benchmark for future adaptation approaches.

C. Models and Techniques

Logistic Regression with L2 Regularization models the log-odds of the binary hit label as a linear function of the feature vector: $\log \frac{p}{1-p} = \mathbf{w}^\top \mathbf{x} + b$. The regularization strength $C = 1/\lambda$ is selected via F1-tuned GridSearchCV. Logistic regression is the primary model throughout this paper because its coefficients are directly interpretable as signed feature importances, enabling cross-genre comparison, and because it performed strongly in closely related prior work [15, 16].

L1 (Lasso) Regularization replaces the squared-norm penalty with an absolute-norm penalty, inducing exact sparsity by driving many coefficients to zero at the optimal solution. This property makes L1 a natural feature-selection tool [9]: features with non-zero coefficients across all 114 genres are treated as domain-invariant predictors, while features active only in a subset are treated as domain-specific.

Random Forests combine an ensemble of decision trees trained on bootstrap samples with random feature subsampling at each split, providing a nonlinear alternative to logistic regression. Feature importance is measured through the mean decrease in Gini impurity across all trees and splits [17]. Comparing RF importances against L1 coefficients allows us to assess whether the two model families agree on which features matter, and whether nonlinearity offers a predictive advantage within genres.

PCA and PLS both reduce the 13-dimensional feature space before logistic regression is applied. PCA constructs orthogonal components that maximize explained variance in $\mathbf{X}$, retaining those needed to reach 90% cumulative variance. PLS instead maximizes the covariance between $\mathbf{X}$ and $\mathbf{y}$, making it a supervised alternative that prioritizes label-relevant signal over total variance [18]. The number of PLS components is tuned from 1 to 6 by macro F1. The key difference between PCA and PLS in this context is that PLS may discard variance in $\mathbf{X}$ that is unrelated to popularity, potentially yielding a more compact and transferable representation.

Decision Threshold Optimization selects the classification threshold by maximizing macro F1 on the validation fold rather than defaulting to 0.5. This is important given the approximately 25%/75% class imbalance within each genre, where the default threshold systematically over-predicts the majority class. The genre-specific optimal threshold from Experiment 1 is carried forward and reused in all cross-genre evaluations in Experiment 2, ensuring a controlled and fair comparison.

IV. Related Work

A. Hit-Song Prediction with Audio Features

Dhanaraj and Logan [1] established hit-song prediction as a supervised binary classification problem using SVM and AdaBoost on acoustic and lyric features. Pachet and Roy [10] challenged this at scale, finding that global features from 32,000 mixed-genre songs barely exceeded weak baselines, a result now interpretable as a consequence of pooling heterogeneous genres. Ni et al. [11] recovered accuracy on UK Top-40 data while showing that feature importance shifts across decades, providing an early example of temporal domain shift. Fan and Casey [7] demonstrated cross-domain transfer failure between Chinese and UK markets, showing that models trained in one market do not generalize to the other. This directly anticipates the genre-as-domain framing used in this paper. Herremans et al. [15] found that logistic regression outperformed other classifiers for dance-hit prediction, providing a methodological precedent for our approach.

Georgieva et al. [16] trained five classical machine learning models on the same Spotify feature set used in this study, with logistic regression reaching approximately 75% accuracy, making it one of the closest baselines to our within-genre experiments. Middlebrook and Sheik [3] scaled the analysis to approximately 24,000 tracks and reported 88% Random Forest accuracy, while Zangerle et al. [12] validated Spotify's descriptors as sufficient for hit prediction. However, neither study examined cross-genre transfer. Raza and Nanath [5] concluded that there is no universal formula for a hit song, a finding that this work investigates through the lens of genre-specific domain shift. Khan et al. [9] reported 95.15% Random Forest accuracy using feature selection, and Dimolitsas et al. [4] achieved 86% accuracy using a pipeline similar to our baseline. Saragih [19] showed that dominant predictors vary across markets, further illustrating domain-specific feature relevance.

The work most closely related to this study is Sebastian Jung and Mayer [6], who identified genre as the strongest predictor of popularity across six categories in a 30,000-song Spotify dataset. However, their work did not measure cross-genre generalization gaps, compare feature importance across genres, or evaluate dimensionality reduction methods, all of which are addressed here at the scale of 114 genres.

B. Domain Shift and Transfer Learning in Music

Choi et al. [13] showed that CNN features trained on music tagging transfer effectively across MIR tasks, establishing an important reference point for audio domain adaptation. Hung et al. [14] extended this idea to low-resource genre classification through speech-model reprogramming, while Abeßer [20] provided a conceptual overview of unsupervised domain adaptation in audio. Although this paper does not directly implement transfer learning methods, it quantifies the magnitude of domain shift that such approaches would need to overcome.

C. Feature Importance and Foundational MIR

Interiano et al. [17] found that audio features provide only modest predictive improvement across 500,000 UK chart songs, while Schedl [21] documented significant popularity differences across Spotify genres, supporting the treatment of genres as distinct domains. Sciandra and Spera [18] applied a Beta GLMM to Spotify popularity scores as a statistically rigorous alternative to standard machine learning pipelines. Salganik et al. [8] showed that identical songs can achieve different outcomes across listener populations because of social influence, establishing a theoretical upper bound on audio-feature predictability. Tzanetakis and Cook [22] and Bertin-Mahieux et al. [23] introduced the feature families and large-scale datasets that later influenced Spotify's audio-feature API. Mauch et al. [24] demonstrated that audio-feature distributions shift across decades in US popular music, providing a temporal analogue of the cross-genre domain shift studied in this work.

V. Approach

A. Dataset

The Spotify Tracks Dataset is sourced from Kaggle.[1] It contains 114,000 tracks across 114 genres, with 1,000 tracks per genre. Each track is associated with metadata, 13 audio features computed by the Spotify API, and a continuous *popularity* score ranging from 0 to 100. The 13 audio descriptors used as predictors are listed in Table I.

Feature	**Type**	**Description**
danceability	float [0,1]	Suitability for dancing
energy	float [0,1]	Intensity and activity
loudness	float (dB)	Overall loudness
speechiness	float [0,1]	Spoken-word content
acousticness	float [0,1]	Acoustic confidence
instrumentalness	float [0,1]	Absence of vocals
liveness	float [0,1]	Live audience presence
valence	float [0,1]	Musical positiveness
tempo	float (BPM)	Beats per minute
duration_ms	int (ms)	Track length
key	int [0,11]	Pitch class
mode	binary	Major (1) / minor (0)
time_signature	int [3,7]	Beats per bar

TABLE I: Audio features used as predictors in all experiments.

B. Data Preparation

One record is removed due to duplication or missing metadata, leaving 113,999 tracks. Of the 89,741 unique track IDs in the raw dataset, 24,259 appear in more than one genre, a duplication rate of 21.28%. These are retained as separate records since each appearance carries a distinct genre label. However, this cross-genre overlap has an important consequence: because the hit label is computed independently per genre using each genre's own 75th percentile, the same track can receive different labels in different genres. For example, *Lovemark* is labeled a hit in *sad* but not in *emo*, *Reach Out* is a hit in *alt-rock* and *funk* but not in *metal*, and *IN MY REMAINS* is a hit in *grunge* but not in *alternative* or *metal*. In total, 3,332 tracks (3.71% of unique tracks) carry conflicting hit labels across genres, providing early evidence of concept drift examined further in Experiment 2.

The intended positive rate is 25%, although ties at the quantile boundary increase the observed per-genre hit ratio to between 25% and 32%, with a mean of approximately 26.5%. All 13 features are standardized to zero mean and unit variance using StandardScaler. Because positive examples represent only 25–32% of tracks per genre, all logistic regression and Random Forest models use *class_weight='balanced'* to compensate for class imbalance.

C. Dataset Analysis

Popularity varies substantially across genres. The five highest mean popularity genres are pop-film (59.3), k-pop (57.0), chill (53.7), sad (52.4), and grunge (49.6), while the five lowest are chicago-house (12.3), detroit-techno (11.2), latin (8.3), romance (3.2), and iranian (2.2). This wide range motivates the use of a genre-relative hit threshold rather than a single global cutoff.

Audio features also differ markedly across genres. Death-metal has the highest mean energy (0.931) while classical has the lowest (0.190); classical conversely has the highest mean acousticness (0.920). Study music leads in instrumentalness (0.790), comedy in speechiness (0.756), and drum-and-bass in tempo (155 BPM). Valence ranges from 0.058 in sleep to 0.815 in salsa, reflecting the contrast between ambient and dance-oriented genres. Among feature correlations, energy and loudness are strongly positively correlated ($r = 0.762$), energy and acousticness strongly negatively correlated ($r = -0.734$), and danceability and valence moderately correlated ($r = 0.477$). These differences motivate treating genres as separate domains rather than pooling them into a single model.

D. Experimental Design

All experiments train and evaluate a separate model for each of the 114 genres using 5-fold stratified cross-validation. Macro-averaged F1 score is used as the primary evaluation metric throughout, chosen over accuracy because the per-genre class distribution is imbalanced at approximately 25% hits to 75% non-hits.

a) Experiment 1: Within-Genre Logistic Regression (Baseline).: The baseline model uses L2-regularized logistic regression. To motivate the choice of F1 as the primary metric, accuracy-tuned and F1-tuned hyperparameter selection are first compared. The final baseline combines F1-tuned regularization strength C selected via GridSearchCV with a genre-specific decision threshold τ^* optimized using the precision-recall curve on out-of-fold predicted probabilities. This produces a per-genre (C^*, τ^*) pair that serves as the reference point for all subsequent experiments.

b) Experiment 2: Cross-Genre Generalization.: Each baseline model from Experiment 1 is retrained on its full genre dataset and then evaluated on all 114 genres using its optimized threshold τ^*. This produces a 114×114 matrix

[1] https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset

of cross-genre F1 scores from which per-genre generalization gaps are computed as the difference between within-genre and mean cross-genre F1.

c) Experiment 3: L1 Regularization and Feature Selection.: L1-regularized logistic regression is trained independently per genre with F1-tuned C values. The resulting coefficient sparsity patterns are analyzed to identify domain-invariant features, defined as features with non-zero coefficients across all 114 genres, and domain-specific features, defined as those zeroed out in at least one genre.

d) Experiment 4: Random Forest Feature Importance.: A Random Forest classifier with 200 trees is trained per genre with *class_weight='balanced'*. Feature importance is measured using mean decrease in Gini impurity and compared against L1 logistic regression coefficients to examine whether the two model families agree on which features drive hit prediction across genres.

e) Experiment 5: PCA with Logistic Regression.: Principal Component Analysis is applied independently within each genre before fitting an L2 logistic regression with F1-tuned C. Components are retained until 90% of cumulative explained variance is reached, yielding a genre-specific component count. Performance is compared against the Experiment 1 baseline.

f) Experiment 6: PLS with Logistic Regression.: Partial Least Squares replaces PCA as a supervised dimensionality-reduction step. The number of components is tuned between 1 and 6 by maximizing macro F1 in an inner cross-validation loop. PLS+LR performance is compared against both the baseline and PCA+LR across all 114 genres.

E. Implementation Details

All experiments are implemented in Python using scikit-learn 1.7.2, pandas 2.3.3, and NumPy 2.3.5. Visualizations are generated using Matplotlib and Seaborn. All stochastic procedures use *random_state=42* and parallel execution is enabled with *n_jobs=-1*. The full code and experimental outputs are publicly available at https://github.com/MehreenRashid1/Hit-Song-Prediction-Across-Genres.

Experiment 1 searches $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$ via GridSearchCV scored on F1, with threshold optimization performed through cross_val_predict and precision_recall_curve. Experiment 3 uses *penalty='l1'* with the *liblinear* solver. Experiment 4 averages Gini importances over 200 trees fitted on the full genre dataset. Experiments 5 and 6 fit PCA and PLS inside each cross-validation fold to avoid data leakage, with PLS component count selected via an inner F1-scoring loop over $n \in \{1, \ldots, 6\}$.

VI. Experimental Evaluation

A. Experiment 1: Within-Genre Logistic Regression

We compare four tuning strategies to motivate the choice of macro-averaged F1 as the primary evaluation metric. Table II reports mean cross-validated metrics across all 114 genres.

Accuracy-tuned and F1-tuned C selection produce nearly identical results, with mean F1 scores of 0.440 and 0.445 respectively, suggesting that the scoring metric has only a modest influence on regularization selection alone. The largest contrast appears when the decision threshold is optimized for accuracy: although mean accuracy increases to 0.741, mean F1 collapses to 0.092 because the model exploits the approximately 25%/75% class imbalance by predicting most tracks as non-hits. ROC-AUC remains stable at 0.622 across all strategies because it is threshold-independent, confirming that the underlying discriminative signal is unchanged. Combining F1-tuned C with genre-specific threshold optimization via the precision-recall curve achieves the highest mean F1 of 0.467 and is adopted as the baseline for all subsequent experiments.

Strategy	Acc.	F1	AUC
Acc-tuned C	0.597	0.440	0.624
Acc-tuned C + acc. threshold	0.741	0.092	0.622
F1-tuned C	0.593	0.445	0.624
F1-tuned C + F1 threshold	0.503	0.467	0.622

TABLE II: Mean cross-validated metrics across 114 genres for each tuning strategy.

Figure 1 shows per-genre baseline F1 scores. Performance ranges from 0.692 for classical to 0.403 for salsa, with a mean of 0.467 and standard deviation of 0.050. The highest-performing genres are classical (0.692), sleep (0.672), hardcore (0.631), k-pop (0.592), and drum-and-bass (0.574). The lowest are salsa (0.403), synth-pop (0.407), reggaeton (0.407), pagode (0.407), and latino (0.411). The strong performance in classical and sleep is attributable to their extreme and consistent audio profiles: classical occupies a narrow region of very low energy (mean 0.190), high acousticness (0.920), and high instrumentalness, creating a clear boundary between popular and unpopular tracks. Sleep similarly occupies an extreme corner of the feature space with near-zero valence (0.058) and very low energy and tempo. By contrast, genres such as salsa, reggaeton, and synth-pop exhibit more homogeneous feature distributions, making it difficult for a linear classifier to identify a meaningful hit/non-hit boundary from audio features alone.

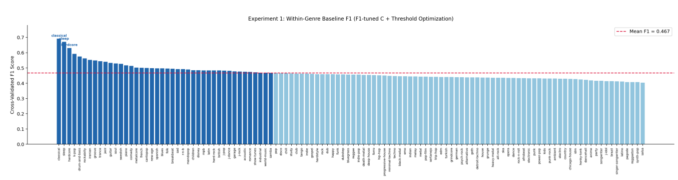

Fig. 1: Within-genre baseline F1 scores across 114 genres sorted in descending order. Darker bars indicate genres above the mean. Performance ranges from 0.692 for classical to 0.403 for salsa.

B. Experiment 2: Cross-Genre Generalization

Each baseline model is retrained on its full genre dataset and evaluated on all 114 genres, producing a 114×114 F1 matrix. Because models are retrained on the full genre data rather than cross-validation folds, the mean within-genre F1 rises slightly to 0.474. Under cross-genre evaluation this drops to 0.352, producing an average generalization gap of 0.114 across 110 of the 114 genres.

Figure 2 shows the per-genre generalization gap. The largest gaps occur for sleep (0.657), iranian (0.527), hardcore (0.402), rockabilly (0.380), and children (0.352), whose narrow and distinctive audio profiles transfer poorly to other genres. In contrast, four genres produce negative gaps: j-idol (−0.004), brazil (−0.005), singer-songwriter (−0.005), and salsa (−0.010), where cross-genre models outperform the within-genre model, likely due to within-genre overfitting on limited positive examples.

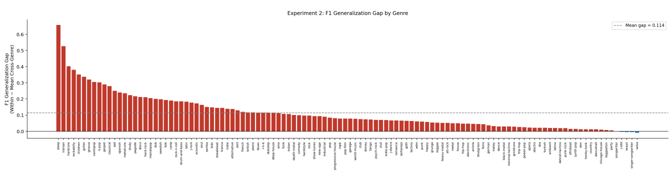

Fig. 2: Per-genre F1 generalization gap computed as within-genre F1 minus mean cross-genre F1. Red bars indicate positive gaps, while blue bars indicate genres where cross-genre performance exceeds within-genre performance.

Table III highlights selected examples from the cross-genre matrix. Three important patterns emerge. First, classical acts as a universal transfer target, appearing in 9 of the top 15 transfer pairs, including from unrelated genres such as disney (0.639) and death-metal (0.617), because its high within-genre separability means many decision boundaries identify its hits by chance. Second, the sleep model produces 222 zero-F1 transfer pairs, as its boundary is calibrated for extremely low-energy tracks. Third, transfer is often asymmetric: metal transfers to hardcore at 0.389 while hardcore transfers to metal at only 0.045, and techno reaches 0.426 on EDM while EDM reaches only 0.303 on techno, suggesting that some genre boundaries subsume others.

Train	**Test**	**F1**	**Note**
disney	classical	0.639	Best pair; unrelated genres
death-metal	classical	0.617	Strong despite dissimilarity
hip-hop	r-n-b	0.463	Culturally related
jazz	classical	0.454	Acoustically similar
metal	hardcore	0.389	Moderate; related genres
hardcore	metal	0.045	Asymmetric reverse
techno	edm	0.426	Similar sonic profile
edm	techno	0.303	Asymmetric reverse
sleep	rock	0.000	Complete failure
children	grindcore	0.000	Complete failure

TABLE III: Selected cross-genre F1 pairs illustrating representative transfer patterns.

The correlation between shared track counts and pairwise cross-genre F1 is $r = 0.093$, indicating that track overlap does not explain transfer performance. The most striking illustration is the singer-songwriter and songwriter pair, which share all 1,000 tracks yet achieve a cross-genre F1 of only 0.413, because the per-genre hit label is derived from each genre's own 75th percentile independently, meaning the same track can be labeled a hit in one genre and a non-hit in the other. These findings confirm that the observed domain shift is driven by concept drift, where the feature-to-label relationship changes across genres, rather than by differences in input distributions alone. A complementary view of the matrix examines which genres produce the most broadly applicable models. The genres whose trained models generalize best across all 114 test genres are chicago-house, turkish, and minimal-techno, each achieving a mean cross-genre F1 above 0.418, while sleep (0.014) and iranian (0.025) produce the least transferable models, consistent with their highly distinctive audio profiles identified earlier.

C. *Experiment 3: L1 Regularization and Feature Selection*

L1-regularized logistic regression is trained independently per genre using F1-tuned C values. The mean cross-validated F1 of 0.443 is below the L2 baseline of 0.467, and L1 outperforms the baseline in only 5 of 114 genres. Although the sparsity penalty slightly reduces predictive performance, it provides insight into which features remain important across genres.

No feature receives a non-zero coefficient in all 114 genres, indicating no fully domain-invariant predictor exists. Figure 3 shows the non-zero count per feature. Valence is the most consistent (104/114), followed by danceability, duration_ms, and mode (100/114 each). Energy and key are the least consistent, receiving zero coefficients in 24 genres each (90/114).

At the genre level, the mean number of active features is 11.0 out of 13, but the distribution is bimodal: 57 genres retain all 13 features while 4 use five or fewer — deep-house (3), salsa (4), honky-tonk (5), and latin (5). A further 26 genres use between 6 and 9 features and 27 use between 10 and 12. Rather than a clean universal subset, the results reveal a continuum from nearly universal predictors such as valence and danceability to highly genre-specific ones such as energy and key.

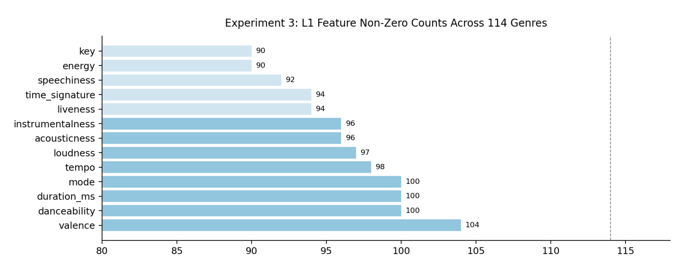

Fig. 3: Number of genres in which each feature receives a non-zero L1 coefficient. No feature is active in all 114 genres.

D. *Experiment 4: Random Forest Feature Importance*

A Random Forest with 200 trees is trained per genre. The mean cross-validated F1 of 0.221 is substantially below both the L2 baseline (0.467) and the L1 model (0.443), with RF outperforming the baseline in only 4 genres. The generally poor RF performance relative to logistic regression can be partly attributed to the limited number of positive training examples per genre: with approximately 250 positive examples available after thresholding, Random Forest has insufficient data to learn reliable multi-feature split combinations without overfitting to noise. A well-regularized linear model is better suited to this regime. Two genres stand out: sleep (0.800) and classical (0.722) substantially outperform both logistic-regression models, likely because the hit/non-hit boundary in these genres is nonlinear yet well-defined by a small number of extreme feature values, so only a few splits are needed to capture it. At the other extreme, salsa (0.022), trip-hop (0.065), dub (0.068), and bluegrass (0.071) perform near random levels.

Figure 4 compares RF Gini importances with L1 consistency. Both agree on the least informative features: key, mode, and time_signature are consistently weak under both measures. Their rankings diverge at the top — RF assigns highest importance to duration_ms (0.103) and loudness (0.100), while L1 consistency favors valence (104/114) and danceability (100/114). This reflects the difference between split-based relevance on training data and the ability to survive regularization across genres.

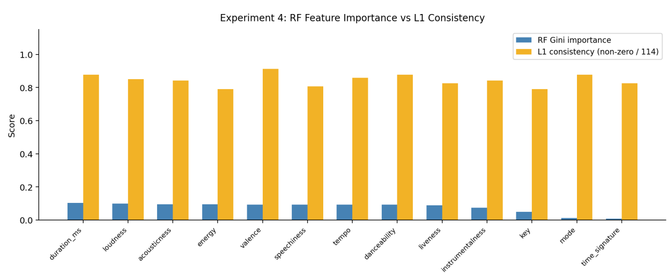

Fig. 4: RF mean Gini importance compared with L1 non-zero consistency per feature.

E. Experiment 5: PCA with Logistic Regression

PCA is applied per genre before logistic regression, retaining components that explain 90% of cumulative variance. This requires a mean of 8.7 components out of 13, with 97 of 114 genres needing 8 or 9, indicating that the audio-feature space is not highly compressible.

The mean F1 of PCA+LR is 0.433, compared with 0.467 for the baseline, a mean decrease of 0.034. PCA underperforms the baseline in 110 of 114 genres. The four exceptions — opera (+0.009), kids (+0.004), techno (+0.004), and british (+0.000) — show negligible gains. The largest losses occur in forro (−0.103), r-n-b (−0.097), mpb (−0.082), and punk (−0.080), suggesting that lower-variance components discarded by PCA still contain label-relevant signal in these genres. Figure 5 shows the per-genre F1 difference.

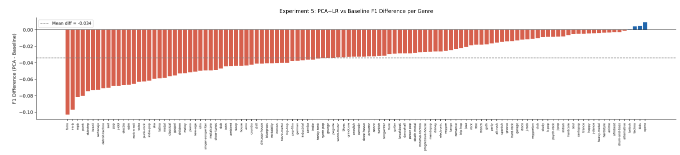

Fig. 5: Per-genre F1 difference between PCA+LR and the baseline, sorted ascending. Red bars indicate decreases; blue bars indicate improvements.

F. Experiment 6: PLS with Logistic Regression

PLS replaces PCA as a supervised dimensionality-reduction step, with components tuned between 1 and 6 by macro F1. The mean F1 of PLS+LR is 0.465, compared with 0.467 for the baseline and 0.433 for PCA+LR. PLS outperforms the baseline in 60 of 114 genres and outperforms PCA+LR in 111. In win counts, PLS wins in 60 genres, the baseline in 54, and PCA in none.

Table IV compares PCA and PLS component counts for selected genres. PLS consistently requires far fewer components: where PCA needs 7–9 to retain 90% of variance, PLS achieves competitive or better F1 with just 1–4. For example, sleep reaches F1 = 0.659 with 4 PLS components versus 7 PCA components (F1 = 0.628), and hardcore achieves 0.633 with 2 PLS components versus 9 PCA components (F1 = 0.624).

Genre	PCA comp.	PLS comp.	Base F1	PCA F1	PLS F1
classical	8	4	0.692	0.635	0.653
sleep	7	4	0.672	0.628	0.659
hardcore	9	2	0.631	0.624	0.633
jazz	9	6	0.541	0.520	0.560
r-n-b	9	3	0.492	0.395	0.475

TABLE IV: PCA and PLS component counts with F1 scores for selected genres.

The PLS component distribution is concentrated at low values: 27 genres use 1 component, 28 use 2, and 21 use 3, meaning 76 of 114 genres are best described by 1–3 supervised components. This contrasts with PCA's mean of 8.7, indicating that label-relevant signal is more concentrated than the overall variance structure suggests. The largest PLS gains over the baseline are in indian (+0.024), happy (+0.023), goth (+0.022), and afrobeat (+0.022); the largest losses are in sertanejo (−0.045), classical (−0.039), and salsa (−0.034). Figure 6 shows the per-genre difference.

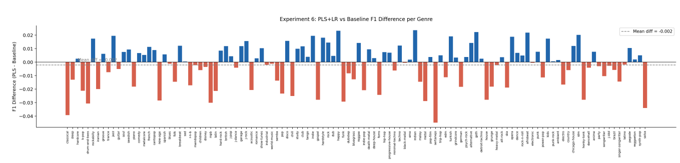

Fig. 6: Per-genre F1 difference between PLS+LR and the baseline. Blue bars indicate improvements; red bars indicate decreases.

G. Research Question Summary

RQ1. Mean within-genre F1 drops from 0.467 to 0.353 under cross-genre evaluation, a gap of 0.114 present in 110 of

114 genres. The remaining four genres (j-idol, brazil, singer-songwriter, salsa) show zero or negative gaps, likely due to within-genre overfitting. The largest gaps occur for sleep (0.657) and iranian (0.527). The correlation between shared tracks and cross-genre F1 is $r = 0.093$, confirming that the shift reflects concept drift rather than distributional differences.

RQ2. Feature relevance varies substantially. No feature has a non-zero L1 coefficient in all 114 genres. Valence is the most consistent (104/114), followed by danceability, duration_ms, and mode (100/114 each). Energy and key are the most genre-specific (90/114). Among genres, 57 retain all 13 features while 4 retain 5 or fewer.

RQ3. L1 does not identify a compact domain-invariant set. No feature survives regularization in all genres, and the mean L1 F1 of 0.443 is below the baseline of 0.467. The results reveal a continuum from near-universal to highly genre-specific features.

RQ4. RF and L1 agree that key, mode, and time_signature are least informative. RF ranks duration_ms and loudness highest while L1 favors valence and danceability. RF achieves a mean F1 of only 0.221, outperforming the baseline in 4 genres, with sleep (0.800) and classical (0.722) as notable exceptions where nonlinear boundaries help.

RQ5. Neither PCA nor PLS consistently improves upon the baseline. PCA underperforms in 110 of 114 genres (mean F1 0.433), with a mean of 8.7 components needed, indicating no compact low-dimensional structure. PLS is more competitive (mean F1 0.465, 60 wins), and 76 genres need only 1–3 components, showing label-relevant signal is concentrated. Neither method surpasses the full-feature baseline overall.

VII. Limitations

Several limitations should be considered when interpreting these results.

Dataset scope. Each genre contains only $\sim$250 positive examples after thresholding, which may contribute to the within-genre overfitting observed in low-signal genres. Popularity scores are Spotify-specific and may not reflect broader commercial success, and the genre taxonomy is platform-defined rather than musicologically rigorous.

Feature representation. The 13 Spotify descriptors are pre-computed high-level features. While sufficient for hit prediction [12], they may discard low-level acoustic detail relevant to certain genres, bounding the analysis to what these features capture.

Hit label construction. Defining hits as the top 25th percentile within each genre produces relative rather than absolute labels, and results are sensitive to this choice.

Model selection and thresholds. Logistic regression is the primary model and more expressive alternatives such as gradient boosting were not evaluated. The cross-genre evaluation also applies within-genre decision thresholds to target genres with potentially different class distributions, which may not be optimal.

VIII. Conclusion

This paper examined whether a universal audio-feature formula for hit songs exists across 114 musical genres. The answer is no. Models trained within a genre achieve a mean F1 of 0.467 but drop to 0.353 when evaluated cross-genre, a gap present in 110 of 114 genres. The four exceptions show near-zero or negative gaps, attributable to within-genre overfitting rather than genuine transferability. The near-zero correlation between shared tracks and cross-genre F1 ($r = 0.093$), together with 3,332 tracks carrying conflicting hit labels across genres, confirms that the shift is driven by concept drift: the same audio features map to different popularity outcomes depending on genre context.

Feature importance analysis reinforces this conclusion. No feature has a non-zero L1 coefficient in all 114 genres, ruling out a compact domain-invariant set. Instead, feature relevance follows a continuum, with valence and danceability most consistent and energy and key most genre-specific. Random Forest and L1 agree that key, mode, and time_signature are least informative globally, but diverge at the top of the importance ranking, reflecting the difference between nonlinear split-based relevance and regularization-based consistency. Random Forest achieves a mean F1 of only 0.221, offering no general advantage over logistic regression except in genres with non-linear boundaries such as sleep and classical. Dimensionality reduction does not recover performance lost to domain shift: PCA underperforms the baseline in 110 genres, and PLS, while more competitive (mean F1 0.465, 60 genre wins), does not surpass the full-feature model overall.

Together, these findings suggest that hit-song prediction is inherently genre-conditioned. Rather than a single global model, effective prediction appears to require genre-aware approaches that account for the distinct audio profiles and popularity dynamics of each domain. Future work could explore domain adaptation methods such as transfer learning or genre-conditioned neural architectures, investigate whether richer representations such as raw audio embeddings reduce the generalization gap, and examine whether temporal drift within genres compounds the cross-genre shift observed here.

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